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AIRFOIL ASSEMBLY

Abstract

An airfoil assembly for a gas turbine engine includes a first blade having a first leading edge and a first trailing edge, and a second blade circumferentially spaced from the first blade where the second blade has a second leading edge and a second trailing edge. The first blade includes a feature formed on the first leading edge where the first blade includes a chord extending from the first leading edge to the first trailing edge. The chord is located radially where a projection of a line normal to an intersection of the chord with the first trailing edge intersects the second leading edge. At least a portion of the feature is radially located at or inboard of the chord.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of U.S. application Ser. No. 17/994,549 filed Nov. 28, 2022, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present subject matter relates generally to components of a gas turbine engine, or more particularly to an airfoil assembly.

BACKGROUND

[0003] A gas turbine engine generally includes a fan and a turbomachine arranged in flow communication with one another. Additionally, the turbomachine of the gas turbine engine generally includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. In operation, air is provided from the fan to an inlet of the compressor section where one or more axial compressors progressively compress the air until it reaches the combustion section. Fuel is mixed with the compressed air and burned within the combustion section to provide combustion gases. The combustion gases are routed from the combustion section to the turbine section. The flow of combustion gases through the turbine section drives the turbine section and is then routed through the exhaust section, e.g., to atmosphere.

[0004] The fan is driven by the turbomachine. The fan includes a plurality of circumferentially spaced fan blades extending radially outward from a rotor disk. Rotation of the fan blades creates an airflow through the inlet to the compressor section of the turbomachine, as well as an airflow over the turbomachine.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0006] FIG. 1 is a schematic, cross-sectional view of an exemplary, unducted gas turbine engine according to various embodiments of the present subject disclosure.

[0007] FIG. 2 is a schematic view of an exemplary airfoil according to various embodiments of the present disclosure.

[0008] FIG. 3A is a schematic view of an exemplary airfoil according to an embodiment of the present disclosure, FIG. 3B is a graph plotting an axial sweep profile of the airfoil of FIG. 3A according to an embodiment of the present disclosure, and FIG. 3C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 3B according to an embodiment of the present disclosure.

[0009] FIG. 4A is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure, FIG. 4B is a graph plotting an axial sweep profile of the airfoil of FIG. 4A according to an embodiment of the present disclosure, and FIG. 4C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 4B according to an embodiment of the present disclosure.

[0010] FIG. 5A is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure, FIG. 5B is a graph plotting an axial sweep profile of the airfoil of FIG. 5A according to an embodiment of the present disclosure, and FIG. 5C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 5B according to an embodiment of the present disclosure.

disclosure.

[0011] FIG. 6A is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure, FIG. 6B is a graph plotting an axial sweep profile of the airfoil of FIG. 6A according to an embodiment of the present disclosure, and FIG. 6C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 6B according to an embodiment of the present disclosure.

[0012] FIG. 7A is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure, FIG. 7B is a graph plotting an axial sweep profile of the airfoil of FIG. 7A according to an embodiment of the present disclosure, and FIG. 7C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 7B according to an embodiment of the present disclosure.

[0013] FIG. 8A is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure, FIG. 8B is a graph plotting an axial sweep profile of the airfoil of FIG. 8A according to an embodiment of the present disclosure, and FIG. 8C is a graph plotting a derivative of the axial sweep profile depicted in FIG. 8B according to an embodiment of the present disclosure.

[0014] FIG. 9A is a schematic diagram depicting exemplary steady pressure fields associated with an exemplary baseline airfoil.

[0015] FIG. 9B is a schematic diagram depicting exemplary steady pressure fields associated with an exemplary airfoil according to embodiments of the present disclosure.

[0016] FIG. 10 is a schematic view of an exemplary airfoil according to another embodiment of the present disclosure.

[0017] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present subject matter.

DETAILED DESCRIPTION

[0018] Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.

[0019] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0020] As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0021] The terms “forward” and “aft” refer to relative positions within a turbomachine, gas turbine engine, or vehicle and refer to the normal operational attitude of the same. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

[0022] The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

[0023] The terms “coupled”, “fixed”, “attached to”, and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

[0024] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise.

[0025] The term “at least one of” in the context of, e.g., “at least one of A, B, and C” refers to only

A, only B, only C, or any combination of A, B, and C.

[0026] Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0027] The term “turbomachine” or “turbomachinery” refers to a machine including one or more compressors, a heat generating section (e.g., a combustion section), and one or more turbines that together generate a torque output.

[0028] The term “gas turbine engine” refers to an engine having a turbomachine as all or a portion of its power source. Example gas turbine engines include turbofan engines, turboprop engines, turbojet engines, turboshaft engines, etc., as well as hybrid-electric versions of one or more of these engines.

[0029] In certain aspects of the present disclosure, an airfoil assembly for a turbomachine is provided. The airfoil assembly generally includes circumferentially spaced airfoils or blades where the blades include a leading edge shape variation feature to reduce the noise effects resulting from pressure waves generated by the blades. For example, embodiments of the present disclosure reduce the pressure signature generated by the rotating blades by blocking or disrupting the pressure waves emanating from the blades by changing the axial sweep profile of the leading edge of the blade.

[0030] Referring now to FIG. 1, a schematic cross-sectional view of a gas turbine engine **100** is provided according to an example embodiment of the present disclosure. Particularly, FIG. 1 provides a turbofan engine having a rotor assembly with a single stage of unducted rotor blades. In such a manner, the rotor assembly may be referred to herein as an “unducted fan,” or the entire gas turbine engine **100** may be referred to as an “unducted turbofan engine.” In addition, the gas turbine engine **100** of FIG. 1 includes a third stream extending from the compressor section to a rotor assembly flowpath over the turbomachine, as will be explained in more detail below.

[0031] For reference, the gas turbine engine **100** defines an axial direction A, a radial direction R, and a circumferential direction C. Moreover, the gas turbine engine **100** defines an axial centerline or longitudinal axis **112** that extends along the axial direction A. In general, the axial direction A extends parallel to the longitudinal axis **112**, the radial direction R extends outward from and inward to the longitudinal axis **112** in a direction orthogonal to the axial direction A, and the circumferential direction extends three hundred sixty degrees (360°) around the longitudinal axis **112**. The gas turbine engine **100** extends between a forward end **114** and an aft end **116**, e.g., along the axial direction A.

[0032] The gas turbine engine **100** includes a turbomachine **120** and a rotor assembly, also referred to as a fan section **150**, positioned upstream thereof. Generally, the turbomachine **120** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. Particularly, as shown in FIG. 1, the turbomachine **120** includes a core cowl **122** that defines an annular core inlet **124**. The core cowl **122** further encloses at least in part a low pressure system and a high pressure system. For example, the core cowl **122** depicted encloses and supports at least in part a booster or low pressure (“LP”) compressor **126** for pressurizing the air that enters the turbomachine **120** through core inlet **124**. A high pressure (“HP”), multi-stage, axial-flow compressor **128** receives pressurized air from the LP compressor **126** and further increases the pressure of the air. The pressurized air stream flows downstream to a combustor **130** of the combustion section where fuel is injected into the pressurized air stream and ignited to raise the temperature and energy level of the pressurized air.

[0033] It will be appreciated that as used herein, the terms “high/low speed” and “high/low pressure” are used with respect to the high pressure/high speed system and low pressure/low speed system interchangeably. Further, it will be appreciated that the terms “high” and “low” are used in this same context to distinguish the two systems, and are not meant to imply any absolute speed

and/or pressure values.

[0034] The high energy combustion products flow from the combustor **130** downstream to a high pressure turbine **132**. The high pressure turbine **132** drives the high pressure compressor **128** through a high pressure shaft **136**. In this regard, the high pressure turbine **132** is drivingly coupled with the high pressure compressor **128**. The high energy combustion products then flow to a low pressure turbine **134**. The low pressure turbine **134** drives the low pressure compressor **126** and components of the fan section **150** through a low pressure shaft **138**. In this regard, the low pressure turbine **134** is drivingly coupled with the low pressure compressor **126** and components of the fan section **150**. The LP shaft **138** is coaxial with the HP shaft **136** in this example embodiment. After driving each of the turbines **132**, **134**, the combustion products exit the turbomachine **120** through a turbomachine exhaust nozzle **140**.

[0035] Accordingly, the turbomachine **120** defines a working gas flowpath or core duct **142** that extends between the core inlet **124** and the turbomachine exhaust nozzle **140**. The core duct **142** is an annular duct positioned generally inward of the core cowl **122** along the radial direction R. The core duct **142** (e.g., the working gas flowpath through the turbomachine **120**) may be referred to as a second stream.

[0036] The fan section **150** includes a fan **152**, which is the primary fan in this example embodiment. For the depicted embodiment of FIG. 1, the fan **152** is an open rotor or unducted fan **152**. In such a manner, the gas turbine engine **100** may be referred to as an open rotor engine.

[0037] As depicted, the fan **152** includes an array of airfoils arranged around the longitudinal axis **112** of engine **100**, and more particularly includes an array of fan blades **154** (only one shown in FIG. 1) arranged around the longitudinal axis **112** of engine **100**. The fan blades **154** are rotatable, e.g., about the longitudinal axis **112**. As noted above, the fan **152** is drivingly coupled with the low pressure turbine **134** via the LP shaft **138**. For the embodiments shown in FIG. 1, the fan **152** is coupled with the LP shaft **138** via a speed reduction gearbox **155**, e.g., in an indirect-drive or geared-drive configuration.

[0038] Moreover, the array of fan blades **154** can be arranged in equal spacing around the longitudinal axis **112**. Each fan blade **154** has a proximal end or root and a distal end or tip and a span defined therebetween. For descriptive purposes, reference will be made to a “tip radius”, represented as “R.sub.tip”, of the fan blade **154**. The tip radius R.sub.tip is the radial distance from the longitudinal axis **112** to the outermost radial coordinate or a tip **157** of the fan blade **154**, typically at the leading edge of the fan blade **154**. A point located at the tip **157** would be referred to as 100% of tip radius R.sub.tip, and a point at the longitudinal axis **112** would be referred to as 0% of tip radius R.sub.tip. Each fan blade **154** defines a pitch change or central blade axis **156**. For this embodiment, each fan blade **154** of the fan **152** is rotatable about its central blade axis **156**, e.g., in unison with one another. One or more actuators **158** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan blades **154** about their respective central blade axes **156**.

[0039] The fan section **150** further includes an array of airfoils positioned aft of the fan blades **154** and also disposed around longitudinal axis **112**, and more particularly includes a fan guide vane array **160** that includes fan guide vanes **162** (only one shown in FIG. 1) disposed around the longitudinal axis **112**. For this embodiment, the fan guide vanes **162** are not rotatable about the longitudinal axis **112**. Each fan guide vane **162** has a proximal end or root and a distal end or tip and a span defined therebetween. The fan guide vanes **162** may be unshrouded as shown in FIG. 1 or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **162** along the radial direction R or attached to the fan guide vanes **162**.

[0040] Each fan guide vane **162** defines a central blade axis **164**. For this embodiment, each fan guide vane **162** of the fan guide vane array **160** is rotatable about its respective central blade axis **164**, e.g., in unison with one another. One or more actuators **166** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan guide vane **162** about its respective

central blade axis **164**. However, in other embodiments, each fan guide vane **162** may be fixed or unable to be pitched about its central blade axis **164**. The fan guide vanes **162** are mounted to a fan cowl **170**.

[0041] As shown in FIG. **1**, in addition to the fan **152**, which is unducted, a ducted fan **184** is included aft of the fan **152**, such that the gas turbine engine **100** includes both a ducted and an unducted fan which both serve to generate thrust through the movement of air without passage through at least a portion of the turbomachine **120** (e.g., without passage through the HP compressor **128** and combustion section for the embodiment depicted). The ducted fan **184** is rotatable about the same axis (e.g., the longitudinal axis **112**) as the fan blade **154**. The ducted fan **184** is, for the embodiment depicted, driven by the low pressure turbine **134** (e.g. coupled to the LP shaft **138**). In the embodiment depicted, as noted above, the fan **152** may be referred to as the primary fan, and the ducted fan **184** may be referred to as a secondary fan. It will be appreciated that these terms “primary” and “secondary” are terms of convenience, and do not imply any particular importance, power, or the like.

[0042] The ducted fan **184** includes a plurality of fan blades (not separately labeled in FIG. **1**) arranged in a single stage, such that the ducted fan **184** may be referred to as a single stage fan. The fan blades of the ducted fan **184** can be arranged in equal spacing around the longitudinal axis **112**. Each blade of the ducted fan **184** has a proximal end or root and a distal end or tip and a span defined therebetween.

[0043] The fan cowl **170** annularly encases at least a portion of the core cowl **122** and is generally positioned outward of at least a portion of the core cowl **122** along the radial direction R. Particularly, a downstream section of the fan cowl **170** extends over a forward portion of the core cowl **122** to define a fan duct flowpath, or simply a fan duct **172**. According to this embodiment, the fan flowpath or fan duct **172** may be understood as forming at least a portion of the third stream of the engine **100**.

[0044] Incoming air may enter through the fan duct **172** through a fan duct inlet **176** and may exit through a fan exhaust nozzle **178** to produce propulsive thrust. The fan duct **172** is an annular duct positioned generally outward of the core duct **142** along the radial direction R. The fan cowl **170** and the core cowl **122** are connected together and supported by a plurality of substantially radially-extending, circumferentially-spaced stationary struts **174** (only one shown in FIG. **1**). The stationary struts **174** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to the stationary struts **174** may be used to connect and support the fan cowl **170** and/or core cowl **122**. In many embodiments, the fan duct **172** and the core duct **142** may at least partially co-extend (generally axially) on opposite sides (e.g., opposite radial sides) of the core cowl **122**. For example, the fan duct **172** and the core duct **142** may each extend directly from a leading edge **144** of the core cowl **122** and may partially co-extend generally axially on opposite radial sides of the core cowl **122**.

[0045] The gas turbine engine **100** also defines or includes an inlet duct **180**. The inlet duct **180** extends between the engine inlet **182** and the core inlet **124**/fan duct inlet **176**. The engine inlet **182** is defined generally at the forward end of the fan cowl **170** and is positioned between the fan **152** and the fan guide vane array **160** along the axial direction A. The inlet duct **180** is an annular duct that is positioned inward of the fan cowl **170** along the radial direction R. Air flowing downstream along the inlet duct **180** is split, not necessarily evenly, into the core duct **142** and the fan duct **172** by a fan duct splitter or leading edge **144** of the core cowl **122**. In the embodiment depicted, the inlet duct **180** is wider than the core duct **142** along the radial direction R. The inlet duct **180** is also wider than the fan duct **172** along the radial direction R.

[0046] Notably, for the embodiment depicted, the engine **100** includes one or more features to increase an efficiency of a third stream thrust, Fn3S (e.g., a thrust generated by an airflow through the fan duct **172** exiting through the fan exhaust nozzle **178**, generated at least in part by the ducted fan **184**). In particular, the engine **100** further includes an array of inlet guide vanes **186** positioned

in the inlet duct **180** upstream of the ducted fan **184** and downstream of the engine inlet **182**. The array of inlet guide vanes **186** are arranged around the longitudinal axis **112**. For this embodiment, the inlet guide vanes **186** are not rotatable about the longitudinal axis **112**. Each inlet guide vanes **186** defines a central blade axis (not labeled for clarity), and is rotatable about its respective central blade axis, e.g., in unison with one another. In such a manner, the inlet guide vanes **186** may be considered a variable geometry component. One or more actuators **188** are provided to facilitate such rotation and therefore may be used to change a pitch of the inlet guide vanes **186** about their respective central blade axes. However, in other embodiments, each inlet guide vanes **186** may be fixed or unable to be pitched about its central blade axis.

[0047] Further, located downstream of the ducted fan **184** and upstream of the fan duct inlet **176**, the gas turbine engine **100** includes an array of outlet guide vanes **190**. As with the array of inlet guide vanes **186**, the array of outlet guide vanes **190** are not rotatable about the longitudinal axis **112**. However, for the embodiment depicted, unlike the array of inlet guide vanes **186**, the array of outlet guide vanes **190** are configured as fixed-pitch outlet guide vanes.

[0048] Further, it will be appreciated that for the embodiment depicted, the fan exhaust nozzle **178** of the fan duct **172** is further configured as a variable geometry exhaust nozzle. In such a manner, the engine **100** includes one or more actuators **192** for modulating the variable geometry exhaust nozzle. For example, the variable geometry exhaust nozzle may be configured to vary a total cross-sectional area (e.g., an area of the nozzle in a plane perpendicular to the longitudinal axis **112**) to modulate an amount of thrust generated based on one or more engine operating conditions (e.g., temperature, pressure, mass flowrate, etc. of an airflow through the fan duct **172**). A fixed geometry exhaust nozzle may also be adopted.

[0049] Moreover, referring still to FIG. **1**, in exemplary embodiments, air passing through the fan duct **172** may be relatively cooler (e.g., lower temperature) than one or more fluids utilized in the turbomachine **120**. In this way, one or more heat exchangers **200** may be positioned in thermal communication with the fan duct **172**. For example, one or more heat exchangers **200** may be disposed within the fan duct **172** and utilized to cool one or more fluids from the core engine with the air passing through the fan duct **172**, as a resource for removing heat from a fluid, e.g., compressor bleed air, oil or fuel.

[0050] Referring now to FIG. **2**, a schematic and fragmentary view of an exemplary airfoil assembly **210** in accordance with various embodiments of the present disclosure is provided. The exemplary airfoil assembly **210** may be configured for use as the fan **152** of the engine **100** as depicted in FIG. **1**. The airfoil assembly **210** includes an array of airfoils or blades **214** (only one shown in FIG. **2**) that are regularly spaced apart circumferentially around a disk or hub of a rotor centered on the longitudinal axis **112** (FIG. **1**) of the fan **152** (FIG. **1**). Each blade **214** includes a proximal end **250** (i.e., an inboard end in the radial direction **R** toward the longitudinal axis **112** (FIG. **1**)) and a distal end or tip **228** such that a span of the blade **214** is defined between the proximal end **250** and the tip **228**. Blade **214** forms an aerodynamic surface extending along the axial direction **A** between a leading edge **234** and a trailing edge **236**. The blade **214** extends outward from the proximal end **250** in the radial direction **R**. In the illustrated embodiment, the leading edge **234** includes an inboard portion **242** that extends outward in the radial direction **R** to a particular span location and an outboard portion **244** that extends from the inboard portion **242** to the tip **228**. In the illustrated embodiment, the leading edge **234** of the inboard portion **242** sweeps forward in the axial direction **A**, and the leading edge **234** of the outboard portion **244** begins sweeping aft in the axial direction **A** outboard of the inboard portion **242**. As will be described in greater detail below, blade **214** includes a feature **240** on the leading edge **234** that disrupts a pressure wave radiated from an adjacent blade **214** to achieve noise reduction (e.g., at cruise condition/operation). In exemplary embodiments of the present disclosure, the feature **240** is a shape variation on or along the leading edge **234** of the blade **214** on the outboard portion **244** of the blade **214** that changes the axial sweep profile of the leading edge **234** of the blade **214**. For

example, as will be described in further detail below, in exemplary embodiments of the present disclosure, the feature **240** provides a decrease in sweep of the leading edge **234** of the blade **214** beginning at a particular radial or span location of the blade **214** and extending to the tip **228** of the blade **214** to provide a single region of negative sweep gradient of the leading edge **234**. For example, in the illustrated embodiment, the leading edge **234** of the blade **214** transitions from sweeping forward (e.g., the inboard portion **242**) to sweeping aft (the outboard portion **244**) at a particular span location **246** (e.g., a span location corresponding to a forward-most axial location of the leading edge **234**). As the leading edge **234** begins sweeping aft outboard of the span location **246** toward the tip **238**, the feature **240** causes a change or deviation to the axial sweep profile such that the feature **240** causes a reduction in the sweep angle or profile along at least a portion of the leading edge **234**. Thus, in exemplary embodiments of the present disclosure, the feature **240** is located in an acoustically active span of the blade **214**. An acoustically active portion or span of the blade **214** may be determined, for example, via a relationship between a source strength distributed radially along the blade **214** and a radiation efficiency along the blade **214**. The acoustically active portion of the blade **214** may be determined by multiplying an acoustic source strength distributed radially along the blade **214** by an acoustic Green's function or radiation efficiency (e.g., the ability of noise sources to propagate acoustic energy to surrounding media) along the blade **214**. The radiation efficiency may be any known relation describing the effective strength of a noise source on the airfoil, fan or propeller blade to an observer location of interest, and may be dependent on the airfoil shape, size, flow conditions, combinations thereof, or the like. In some exemplary embodiments, the trailing edge **236** of the blade **214** is configured having a smooth, curved profile (e.g., without steps or abrupt axial sweep changes/transitions).

[0051] Referring now to FIG. 3A-8C, FIGS. 3A, 4A, 5A, 6A, 7A, and 8A depict schematic views of exemplary airfoils or blades **214** according to various embodiments of the present disclosure. FIGS. 3B, 4B, 5B, 6B, 7B, and 8B depict graphs plotting an axial sweep profile of the leading edge **234** of the airfoils or blades **214** depicted in respective FIGS. 3A, 4A, 5A, 6A, 7A, and 8A where the axial sweep profile depicts a change in axial coordinates Z of the leading edge **234** relative to or as a function of a change in radial coordinates R or span of the leading edge **234**. FIGS. 3C, 4C, 5C, 6C, 7C, and 8C depict graphs plotting a derivative of the axial sweep profile depicted in respective FIGS. 3B, 4B, 5B, 6B, 7B, and 8B.

[0052] Referring specifically to FIG. 3A, an exemplary embodiment of an airfoil or blade **214** is depicted where a shape or profile of the blade **214** is defined by a vertical axis representing a percentage of the span of the blade **214** (e.g., extending radially from, for example, a proximal end **250** of the blade **214** to a distal end or tip **228** of the blade **214**) and a horizontal or Z -axis representing the profile of the blade **214** relative to the central blade or pitch change axis **156** (FIG. 1) of the blade **214**. For example, in FIG. 3A, $Z=0$ represents the central blade or pitch change axis **156** (FIG. 1) of the blade **214** where a forward location relative to the central blade or pitch change axis **156** (FIG. 1) is right-to-left in FIG. 3A and an aft location relative to the central blade or pitch change axis **156** (FIG. 1) is left-to-right in FIG. 3A.

[0053] In FIG. 3A, the blade **214** includes a leading edge **234** and a trailing edge **236**. As depicted in FIG. 3A, the blade **214** also includes a feature **240** located on the leading edge **234**. In FIG. 3A, a baseline profile of the leading edge **234** of the blade **214** without the feature **240** is defined by the baseline profile line **252**. As illustrated in FIG. 3A, the feature **240** includes an inboard portion **260** and an outboard portion **262**. In the embodiment illustrated in FIG. 3A, the feature **240** provides a single non-monotonic sweep of the leading edge **234** located between a medial location or midsection **264** of the blade **214** and the tip **228** of the blade **214**. In the illustrated embodiment, the inboard portion **260** of the feature **240** begins at or outboard of 80% of the span of the blade **214**, and the outboard portion **262** of the feature **240** extends from the inboard portion **260** to the tip **228** of the blade **214**.

[0054] In the illustrated embodiment, the inboard portion **260** of the feature **240** effectively reduces

the sweep of the leading edge **234** of the blade **214** along at least a portion of the span of the leading edge **234**. FIG. 3B depicts a graph plotting an axial sweep profile of the leading edge **234** of the blade **214** of FIG. 3A in degrees. For example, in FIG. 3B, the vertical axis represents a percentage of the span of the leading edge **234** of the blade **214** (e.g., extending radially from, for example, the proximal end **250** of the blade **214** to the distal end or tip **228** of the blade **214**) and the horizontal axis represents the change in a Z-position of the leading edge **234** of the blade **214** relative to the change in the span of the leading edge **234** of the blade **214**. The axial sweep profile of the leading edge **234** may be represented by: $[0055] \frac{dZ_{\text{sub.LE}}}{dR}$

where $dZ_{\text{sub.LE}}$ represents a change in Z-position of the leading edge **234**, and dR represents a change in the span represented by the radial location R. As depicted in FIGS. 3A and 3B, the inboard portion **260** of the feature **240** results in a lower or reduced sweep (e.g., a negative change in Z relative to a change in the span or radial location R of the leading edge **234**) while the outboard portion **262** results in a greater rate of change to Z relative to the radial coordinate R of the leading edge **234** of the blade **214**. FIG. 3C depicts a graph plotting a derivative of the axial sweep profile depicted in FIG. 3B. The derivative of the axial sweep profile is represented as: $[0056] \frac{d(\text{sup.}2Z_{\text{sub.LE}})}{dR_{\text{sup.}2}}$

As illustrated in FIG. 3C, the feature **240** on the leading edge **234** of the blade **214** depicted in FIG. 3A results in a single region of negative sweep gradient **270** (e.g., a decreasing sweep profile). As shown in FIGS. 3A and 3B, the blade **214** has a high level of sweep at the leading edge **234** near the distal end or tip **228** of the blade **214**. This high level of sweep enables the blade **214** to achieve high levels of aerodynamic performance with reduced levels of noise from pressure waves radiated from the blade **214** when the blade **214** is rotating. In the embodiment shown, the axial sweep profile of the blade **214** is maintained in the outboard portion **262** of the feature **240**. However, the leading edge **234** of the blade **214** containing the feature **240** is closer to the pitch axis **156** (FIG. 1) at $Z=0$, thereby reducing the twisting moment of the blade **214** about the pitch axis **156** (FIG. 1) due to centrifugal and aerodynamic loads. Thus, the feature **240** allows for increased aerodynamic performance and lower noise while reducing twisting moments and induced mechanical stresses at the root of the blade **214**.

[0057] FIGS. 4A-4C, 5A-5C, 6A-6C, 7A-7C, and 8A-8C depict axial sweep profile graph plots, and axial sweep profile derivative graph plots of various different embodiments of a blade **214** with a feature **240** on the leading edge **234** of the blade **214** according to embodiments of the present disclosure. In the illustrated embodiments, an inboard portion **260** of the feature **240** begins at varying span or radial locations R and has varying profiles in the Z-direction while maintaining a reduced sweep profile and a single region of negative sweep gradient. In each of the illustrated embodiments, the inboard portion **260** begins at a span or radial location R between a medial location or midsection **264** of the blade **214** and the tip **228** of the blade **214**, and an outboard portion **262** of the feature **240** extends radially outward from the inboard portion **260** to the tip **228** of the blade **214**.

[0058] Additionally in the embodiments shown in FIGS. 5A-5C and 6A-6C, the outboard portion **262** of the feature **240** has an increased sweep near the tip **228** of the blade **214** relative to the baseline profile line **252** of a blade without feature **240**. This increased sweep improves the aerodynamic performance and reduces the noise generated by the blade **214** relative to a blade without the feature **240**. Further, the axial sweep profile can be defined by different types of curves including a piecewise line (FIG. 4B) or a smooth line (FIG. 5B). The feature **240** can be located near the tip **228** (FIGS. 3A, 4A, 5A) (e.g., located at or greater than eighty percent (**80%**) of the span of the blade **214**), more inboard near the medial location **264** (FIGS. 6A, 7A and 8A), or at any radial or span location in between. Additionally, the change in axial sweep of the feature **240** can have varying amplitudes (e.g., the axial sweep of the feature **240** can be moderate (FIG. 8A), or the change in axial sweep of the feature **240** can be more pronounced (FIG. 7A)).

[0059] FIG. 9A is a schematic diagram depicting a steady pressure field in a blade-to-blade

direction associated with an airfoil assembly **290** without a feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A) of the present disclosure, and FIG. 9B is a schematic diagram depicting a steady pressure field in a blade-to-blade direction associated with an airfoil assembly **210** containing a feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A) according to embodiments of the present disclosure. In FIG. 9A, two adjacent blades **292** (e.g., blades **292A** and **292B**) are depicted of the airfoil assembly **290** where each of blades **292A** and **292B** includes a leading edge **294** and a trailing edge **296**. The blades **292A** and **292B** may be configured similar to the blades **214** except without the feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A). In FIG. 9B, two adjacent blades **214** (e.g., blades **214A** and **214B**) are depicted of the airfoil assembly **210** where each of the blades **214A** and **214B** includes a leading edge **234** and a trailing edge **236**, and where the blades **214A** and **214B** are configured with the feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A).

[0060] This circumferential variation of the pressure field generates noise radiating away from a blade **214** as the blade **214** rotates. Referring to FIG. 9B, the feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A) is radially or spanwise located on the leading edge **234** to at least partially block or disrupt a pressure field radiated by an adjacent blade **214**. For example, in some embodiments, the airfoil assembly **210** is configured to reduce noise caused by pressure waves impacting the fuselage of an aircraft that are generated by the blades **214**. In some embodiments, the noise reduction is achieved during a cruise phase of operation of the gas turbine engine **100** (FIG. 1). Thus, in this embodiment, the orientation of the blades **214** is considered to be at a cruise phase position (e.g., via actuators **158** (FIG. 1)). However, it should be understood that embodiments of the present disclosure are applicable to other operational phases of flight and corresponding blade **214** pitch positions.

[0061] The feature **240** is located radially or spanwise on the leading edge **234** of the blade **214** to enable the feature **240** to at least partially block or disrupt a pressure field radiated by an adjacent blade **214**. In some embodiments, a radial or spanwise location the feature **240** is such that at least part of the feature **240** is located at or inboard of a chord **280** defined on the blade **214** where a projection of a line **286** normal (e.g., in a two-dimensional unwrapped view) to the chord **280** at its intersection with the trailing edge **236** intersects the leading edge **234** of the adjacent blade **214**. In FIGS. 9A and 9B, a high pressure zone **282** and a low pressure zone **284** are depicted in connection with airfoil assemblies **290** and **210**, respectively. As indicated by comparing the high pressure zone **282** and low pressure zone **284** between FIGS. 9A and 9B, the presence of the feature **240** on the blades **214** has a blocking effect on the pressure field generated by the adjacent blade **214**, thereby resulting in a reduced noise generation by the blades **214**. For example, as illustrated in FIGS. 9A and 9B, low pressure zones **284** are depicted on the suction side of respective blades **292** and **214**, and high pressure zones **282** are depicted at the leading edges **294** and **234** of respective blades **292** and **214**. As best illustrated in FIG. 9B, the feature **240** (FIG. 2) on blade **214A** blocks the louder noise regions generated by the low pressure region **284** on the blade **214B**.

[0062] Generally, an open or unducted fan architecture can generate high levels of tonal noise at certain frequencies resulting in high levels of tonal noise inside an aircraft cabin (e.g., resulting from the pressure waves impacting the fuselage of the aircraft). Embodiments of the present disclosure reduce the pressure signature generated by the blades by at least partially blocking or disrupting the pressure waves emanating from the blades. By introducing a shape variation on the leading edge of the blade (e.g., the feature **240** (FIGS. 2, 3A, 4A, 5A, 6A, 7A, and 8A)) that changes the axial sweep profile at a particular radial or span location, the shape variation reduces noise while reducing mechanical risk of the blade **214** (e.g., as opposed to increasing the sweep and dihedral of a blade that may introduce increased load at the blade root and/or pitch change mechanism due to increased mass of the blade away from the pitch change axis).

[0063] Referring to FIG. 10, FIG. 10 is a schematic view of an exemplary airfoil or blade **322** of an airfoil assembly **320** according to another embodiment of the present disclosure. The blade **322** and airfoil assembly **320** may be configured similarly to the blade **214** and airfoil assembly **210** of

FIGS. 2-8C except the blade **322** includes a sculpted trailing edge feature **336**. For example, in the illustrated embodiment, blade **322** includes a proximal end **324** (i.e., an inboard end in the radial direction R toward the longitudinal axis **112** (FIG. 1)) and a distal end or tip **326** such that a span of the blade **322** is defined between the proximal end **324** and the tip **326**. Blade **322** forms an aerodynamic surface extending along the axial direction A between a leading edge **328** and a trailing edge **330**. In the illustrated embodiment, the blade **322** includes at its trailing edge **330** the sculpted trailing edge feature **336** (e.g., a wavy feature or plurality of features) configured to facilitate wake mixing to reduce interaction noise caused by the blade **322** wakes impinging on downstream stationary airfoils or stators, as described in U.S. Pat. No. 8,083,487 B2 which is hereby incorporated by reference in its entirety. A baseline **334** trailing edge having a smooth profile is depicted to further illustrate the sculpted trailing edge feature **336**.

[0064] Thus, embodiments of the present disclosure include circumferentially spaced airfoils or blades where the blades include a leading edge shape variation feature to reduce the noise effects resulting from pressure waves generated by the blades. For example, embodiments of the present disclosure reduce the pressure signature generated by the rotating blades by blocking or disrupting the pressure waves emanating from the blades by changing the axial sweep profile of the leading edge of the blade. The shape variation feature includes a non-monotonic variation in sweep and has a portion providing a reduction in sweep on the leading edge at a defined radial location.

[0065] This written description uses examples to disclose the disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0066] While this disclosure has been described as having exemplary designs, the present disclosure can be further modified within the scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

[0067] Further aspects are provided by the subject matter of the following clauses:

[0068] An airfoil assembly for a gas turbine engine, the airfoil assembly comprising: a first blade having a first leading edge and a first trailing edge; and a second blade circumferentially spaced from the first blade, the second blade having a second leading edge and a second trailing edge; and wherein the first blade includes a feature formed on the first leading edge, wherein the first blade includes a chord extending from the first leading edge to the first trailing edge, the chord located radially where a projection of a line normal to an intersection of the chord with the first trailing edge intersects the second leading edge, and wherein at least a portion of the feature is radially located at or inboard of the chord.

[0069] The airfoil assembly of the preceding clause, wherein the feature includes an inboard portion and an outboard portion, and wherein at least part of the inboard portion is radially located at or inboard of the chord.

[0070] The airfoil assembly of any preceding clause, wherein the outboard portion extends to a radial location outboard of the chord.

[0071] The airfoil assembly of any preceding clause, wherein the feature comprises a non-monotonic variation in sweep.

[0072] The airfoil assembly of any preceding clause, wherein the first blade includes a proximal end and a tip, and wherein the feature includes an inboard portion and an outboard portion, and

wherein the outboard portion extends to the tip.

[0073] The airfoil assembly of any preceding clause, wherein the first blade includes a proximal end, a midsection, and a tip, and wherein the feature is located between the midsection and the tip.

[0074] The airfoil assembly of any preceding clause, wherein the feature includes an inboard portion and an outboard portion, and wherein the inboard portion is a reduction in a sweep of the first leading edge.

[0075] The airfoil assembly of any preceding clause, wherein the feature is located in an acoustically active span of the first blade.

[0076] The airfoil assembly of any preceding clause, wherein the first trailing edge comprises a smooth, curved axial profile.

[0077] The airfoil assembly of any preceding clause, wherein the first trailing edge comprises a sculpted trailing edge feature.

[0078] The airfoil assembly of any preceding clause, wherein the first trailing edge incorporates a plurality of spaced-apart wave-shaped projections configured to facilitate wake mixing to reduce interaction noise caused by the blade wakes impinging on downstream stationary blades.

[0079] An airfoil assembly for a gas turbine engine, the airfoil assembly comprising: a plurality of circumferentially spaced blades, wherein each blade of the plurality of blades includes: a leading edge and a trailing edge; a pitch change axis; and a feature formed on the leading edge; and wherein the feature is formed at a radial location R and having a leading edge location Z relative to the pitch change axis, and wherein the feature is defined having a single portion where $d.\sup.2Z/dR.\sup.2 < 0$.

[0080] The airfoil assembly of any preceding clause, wherein each blade of the plurality of blades includes a proximal end, a midsection, and a tip, and wherein the feature is located between the midsection and the tip.

[0081] The airfoil assembly of any preceding clause, wherein the feature comprises a non-monotonic variation in sweep.

[0082] The airfoil assembly of any preceding clause, wherein each blade of the plurality of blades includes a proximal end and a tip, and wherein the feature includes an inboard portion and an outboard portion, and wherein the outboard portion extends to the tip.

[0083] The airfoil assembly of any preceding clause, wherein the plurality of blades includes a first blade and a second blade, and wherein the feature of the first blade disrupts a pressure field and noise radiated by the second blade.

[0084] The airfoil assembly of any preceding clause, wherein the feature includes a reduced axial sweep profile.

[0085] The airfoil assembly of any preceding clause, wherein the feature includes an inboard portion and an outboard portion, and wherein the inboard portion includes a reduced axial sweep profile.

[0086] The airfoil assembly of any preceding clause, wherein the feature includes an inboard portion located at or greater than eighty percent of a span of a respective blade.

[0087] A gas turbine engine, comprising: a turbomachine; and a fan assembly rotatable by the turbomachine, the fan assembly including an airfoil assembly comprising: a first blade having a first leading edge and a first trailing edge; and a second blade circumferentially spaced from the first blade, the second blade having a second leading edge and a second trailing edge; and wherein the first blade includes a feature formed on the first leading edge, wherein a chord is defined on the first blade extending from the first leading edge to the first trailing edge, the chord located radially where a projection of a line normal to an intersection of the chord with the first trailing edge intersects the second leading edge, and wherein at least a portion of the feature is radially located at or inboard of the chord.

[0088] The gas turbine engine of any preceding clause, wherein the fan assembly comprises an unducted fan.

[0089] The gas turbine engine of any preceding clause, wherein the first blade includes a pitch change axis, and wherein the feature is formed at a radial location R and having a leading edge location Z relative to the pitch change axis, and wherein the feature is defined having a single portion where $d.\sup.2Z/dR.\sup.2 < 0$.

[0090] The gas turbine engine of any preceding clause, wherein the feature comprises a non-monotonic variation in sweep.

[0091] The gas turbine engine of any preceding clause, wherein the feature includes an inboard portion and an outboard portion, and wherein the inboard portion is radially located at or inboard of the chord.

[0092] The gas turbine engine of any preceding clause, wherein the first blade includes a proximal end and a tip, and wherein the feature includes an inboard portion and an outboard portion, and wherein the outboard portion extends to the tip.

[0093] The gas turbine engine of any preceding clause, wherein the first blade includes a proximal end, a midsection, and a tip, and wherein the feature is located between the midsection and the tip.

[0094] The airfoil assembly of any preceding clause, wherein the trailing edge comprises a smooth, curved axial profile.

Claims

1. An airfoil assembly, comprising: a plurality of airfoils spaced circumferentially apart about a longitudinal axis, each airfoil of the plurality of airfoils having spaced-apart pressure and suction sides extending radially in span from a root to a tip, and extending axially in chord between spaced apart leading and trailing edges; and wherein at least one airfoil of the plurality of airfoils includes an inboard portion and an outboard portion wherein the outboard portion is located radially outward from the inboard portion, and wherein the leading edge of the inboard portion of the at least one airfoil sweeps forward in an axial direction, and wherein the leading edge of the at least one airfoil sweeps aft in the axial direction outboard of the inboard portion, and wherein the leading edge of the at least one airfoil includes a feature, and wherein at least a portion of the feature reduces a sweep angle along at least a portion of the leading edge of the at least one airfoil in the outboard portion, and wherein the feature extends radially outboard to the tip of the at least one airfoil.

2. The airfoil assembly of claim 1, wherein the feature includes an inboard portion and an outboard portion, and wherein at least part of the inboard portion is radially located at or outboard of a midsection of the at least one airfoil.

3. The airfoil assembly of claim 1, wherein the feature is located at a radial location R, and wherein the radial location R is located at or outboard of eighty percent of a span of the at least one airfoil.

4. The airfoil assembly of claim 1, wherein the feature comprises a non-monotonic variation in sweep.

5. The airfoil assembly of claim 1, wherein the trailing edge of the at least one airfoil comprises a sculpted trailing edge feature.

6. The airfoil assembly of claim 1, wherein the feature is located at a radial location R, and wherein the radial location R is located outboard of a forward-most axial location of the leading edge of the at least one airfoil.

7. The airfoil assembly of claim 1, wherein the feature defines a single region of a negative sweep gradient in the outboard portion of the at least one airfoil.

8. An airfoil assembly, comprising: a plurality of circumferentially spaced airfoils, wherein at least one airfoil of the plurality of airfoils includes: a leading edge and a trailing edge; and a feature formed on the leading edge; and wherein the feature is formed at a radial location R and having a leading edge location Z.sub.LE, and wherein the feature is defined having a single portion where $d.\sup.2Z.\text{sub.LE}/dR.\sup.2 < 0$.

- 9.** The airfoil assembly of claim 8, wherein the at least one airfoil includes a proximal end, a midsection, and a tip, and wherein the feature is located between the midsection and the tip.
- 10.** The airfoil assembly of claim 8, wherein the feature comprises a non-monotonic variation in sweep.
- 11.** The airfoil assembly of claim 8, wherein the at least one airfoil includes a proximal end and a tip, and wherein the feature includes an inboard portion and an outboard portion, and wherein the outboard portion extends to the tip.
- 12.** The airfoil assembly of claim 8, wherein the at least one airfoil is one or more of a guide vane or a blade.
- 13.** The airfoil assembly of claim 8, wherein the feature includes a reduced axial sweep profile.
- 14.** The airfoil assembly of claim 8, wherein the at least one airfoil comprises a first airfoil and second airfoil, and when the first and second airfoil are oriented in a cruise phase position, and wherein the radial location R of at least a portion of the feature on the first airfoil is located at or inboard of a chord a chord defined on the first airfoil where a projection of a line normal to an intersection of the chord with the trailing edge of the first airfoil intersects the leading edge of the second airfoil.
- 15.** The airfoil assembly of claim 8, wherein the trailing edge comprises a sculpted trailing edge feature.
- 16.** The airfoil assembly of claim 8, wherein the feature includes an inboard portion and an outboard portion, and wherein the inboard portion includes a reduced axial sweep profile.
- 17.** The airfoil assembly of claim 8, wherein the feature includes an inboard portion located at or greater than eighty percent of a span of a respective airfoil.
- 18.** An engine, comprising: a fan assembly rotatable about a longitudinal axis; and an airfoil assembly comprising a plurality of airfoils spaced circumferentially apart about the longitudinal axis, wherein the plurality of airfoils comprises at least one airfoil having spaced-apart pressure and suction sides extending radially in span from a root to a tip, and extending axially in chord between spaced apart leading and trailing edges and wherein the at least one airfoil includes an inboard portion and an outboard portion wherein the outboard portion is located radially outward from the inboard portion, and wherein the leading edge of the inboard portion of the at least one airfoil sweeps forward in an axial direction, and wherein the leading edge of the at least one airfoil sweeps aft in the axial direction outboard of the inboard portion, and wherein the at least one airfoil includes a feature that reduces a sweep angle along at least a portion of the leading edge of the at least one airfoil in the outboard portion, and wherein the feature extends radially outboard to the tip of the at least one airfoil.
- 19.** The engine of claim 18, wherein the feature comprises a non-monotonic variation in sweep.
- 20.** The engine of claim 18, wherein the feature is located between a midsection of the at least one airfoil and the tip.
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